

CHAPTER 5
**DETECTION OF ALZHEIMER'S DISEASE
THROUGH AI-DRIVEN AND METHYLATION
DIFFERENCE REGION ANALYSIS OF SIGNIFICANT
EPIGENETIC MODIFICATIONS IN APP, PSEN1,
PSEN2, APOE, MAPT, AND TREM2 GENES**

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INTRODUCTION

Alzheimer's Disease (AD) is a progressive neurological disorder that primarily affects memory, thinking, and behavior (Kamatham et al., 2024). In AD, there are several factors that influence the susceptibility towards this disease. This was evident as the risk for AD was said to increase with age, family records of this disease, unhealthy lifestyle and dietary habits, head trauma and even genetics (Breijyek & Karaman, 2020). Genetics is the most crucial factor that plays a role in the development of AD as genetic factors were known to be involved in at least 70% AD cases (Silva et al., 2019). The inheritance of AD involved two categories of genes, deterministic genes and risk genes ; Deterministic genes are genes that follow the pattern of Mendelian inheritance and can confidently guarantee the development of the disease inherited in most cases (Harden et al., 2022). In the case of AD, mutations in APP and PSEN1/PSEN2 guarantee a 77% pathogenicity of early-onset familial AD (Lanoiselée et al., 2017). Additionally, risk genes are genes that can increase the susceptibility of the disease without absolute guarantee of the disease development, this category is generally impacted by other factors such as, lifestyle, environment and even other genes. Thus, polymorphisms in APOE such as APOE $\epsilon 4$ were said to increase the susceptibility of late-onset AD cases (Polsinelli et al., 2023).

Epigenetic changes have played an important role in the development and progression of Alzheimer's Disease, affecting several key genes in the pathology (Behl et al., 2024). These modifications, which alter the gene expression without changing the DNA sequence can help uncover the complex mechanisms underlying Alzheimer's disease and are able to be used for early detection and intervention (Alkhamash & Alotaibi, 2025). APP gene or the Amyloid Precursor Protein gene can be considered as the central of Alzheimer's disease pathology, as it encodes for the Amyloid Precursor Protein (APP) (Hampel et al, 2021). While the function of the protein itself is unknown, scientists hypothesize that it may bind to other proteins on the surface of cells or help cells attach to one another which helps direct nerve cell movement (Orobets & Karamyshev, 2023). PSEN1 and PSEN2 genes encode the components of the gamma-secretase complex, which is involved in APP processing (Valdes et al., 2025).

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Some studies suggest that PSEN1 mutations influence the methylation of other AD-related genes such as MAPT while PSEN2 expression is downregulated in specific brain regions of AD patients due to epigenetic modifications (De Plano et al., 2024; Xiao et al., 2020). The APOE gene, particularly the $\epsilon 4$ allele, is the strongest genetic risk factor for late-onset AD. Studies stated that the $\epsilon 4$ variant introduces an extra CpG site due to specific single nucleotide polymorphisms (SNPs) (Giallongo et al., 2022). This can increase methylation, potentially silencing or altering APOE expression which affects the amyloid-beta ($A\beta$) clearance, lipid transport, and neuronal health (Lozupone et al., 2023). The MAPT gene encodes the tau protein production and can be influenced by epigenetic modifications. Silencing epigenetic marks (e.g., methylation) may reduce tau levels, potentially protecting against abnormal tau aggregation while activating marks (e.g., histone acetylation) could increase tau production, raising the risk of aggregation and neurodegeneration (Yu et al., 2019). TREM2 gene encodes the protein found on microglia. Protein helps microglia clear out harmful debris, like amyloid plaques, which are sticky protein clumps linked to Alzheimer's disease (Lue et al., 2015). If the TREM2 gene is methylated, it might be silenced, leading to less TREM2 protein. This could weaken the microglia's ability to clean up plaques, worsening disease progression.

With the progress of research, validation of significant known methylation and genetic targets were widely studied in order to understand gene expression with respect towards methylation rates that may have a critical impact towards the downstream pathway signalling related to Alzheimer disease. This was evident as highly significant gene targets such as, APP, PSEN1, PSEN2, APOE, MAPT, TREM2 are often further researched to observe the significance and relevance of this gene with respect towards new gene targets (Neuner et al., 2020; Vega et al., 2022). Furthermore with the progress of technological advancements, AI (Artificial Intelligence) has contributed greatly towards gene-based diagnostics of Alzheimer's biomarkers (Winchester et al. 2023). Whereas, these AI models relied on deep learning workflows such as multi-stage deep learning, support vector machine learning, generalized machine learning, and random forest. They were used to analyze a variety of genomic and methylation data.

This approach allowed for the identification of specific genomic and epigenetic factors that influence the pathogenesis of Alzheimer's disease (Jo et al., 2025; Bahado-Singh et al., 2022). Thus, in order to further AI development and detection of Alzheimer's disease, AI-driven analysis and DMR analysis will be used concurrently in this study to understand key methylation factors significantly expressed in the target genes APP, PSEN1, PSEN2, APOE, MAPT, TREM2.

1. METHODS

Data Collection for DMR Analysis and KEGG Pathway Analysis

The dataset used for DMR analysis was collected from the Gene Expression Omnibus (GEO) with the accession code of GSE244352. This data contained methylation data of genomic DNA taken from the peripheral blood samples of 12 clinically diagnosed Alzheimer's patients and 12 subjects with normal cognition. In order to extract the genomic data, Maxwell RSC Instrument (Promega, USA) with Maxwell RSC Buffy Coat DNA Kit (Promega, USA) was used and further methylation profiling was performed with the use of Illumina HiSeq 2500. This dataset is referenced from the study conducted in "Identification of diagnostic DNA methylation markers in the blood of Japanese Alzheimer's disease patients using methylation capture sequencing" (Mitsumori et al., 2025).

Data Collection for AI-Driven Analysis of RNA-seq Data

Additionally, datasets such as GSE48350 and GSE11882 were also collected from the Gene Expression Omnibus. Both of these datasets contained microarray gene expression profiles on four different brain regions, such as the hippocampus (HC), entorhinal cortex (EC), superior frontal gyrus (SG), and postcentral gyrus (PCG). Postmortem brain tissue was collected from the ADRC brain banks. The total RNA was then extracted from frozen, unfixed tissue using TRIzol reagent (Invitrogen, Carlsbad, CA) and purified using quick spin columns (Qiagen, Valencia, CA). A UV spectrophotometer and Agilent BioAnalyzer were used to measure RNA quality for sharp peaks and 28S/18S RNA.

AI/ML Implementation

To elucidate the role of epigenetic modifications in AD, multilayer perceptron models will be employed with publicly available transcriptomic datasets GSE48350 and GSE11882 from GEO. These models will be utilized to detect the status of a patient's well-being of developing AD based on the selected gene's expression. Furthermore, the analysis will incorporate relevant factors such as sex, age, and Braak stage (stage 0 - II were classified as Normal, stage III-VI were considered indicative of AD pathology) to comprehensively assess their impact on the AD-related genes.

Data Extraction and Preprocessing

The datasets were extracted using GEOparse for fast and efficient loading in the Google Colab environment. Then, the expression data and metadata contained in the dataset were defined and extracted. After that, some columns inside the metadata were dropped as they were unnecessary to be used in the development of the machine learning model. Finally, both of these separate datasets that were already preprocessed and cleaned were combined into one using the concat feature.

Feature Construction and Integration

The set of targets of AD related genes selected were processed further. Each of the genes were specified with associated Affymetrix probe IDs and the average expression value across probes was also calculated to obtain gene-level expression. These genes were chosen and combined with standardized demographic features. To guarantee compatibility with the model, the brain region variable was one-hot encoded. StandardScaler was used to standardize all numerical features, and the resulting matrix was split into training and testing sets (80:20 ratio) with the train_test_split function.

Multilayer Perceptron (MLP) Model Construction and Evaluation

First, the target was defined, which holds a series of 0s and 1s, identified as normal and AD, respectively, corresponding to the disease status of each sample in the same order as the rows of the features in the dataset.

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Then, the data was split into `x_train`, `y_train`, `x_test`, and `y_test` sets. The targets were then converted into Numpy arrays for compatibility and to optimize performance for training a Keras neural network. Lastly, the model was designed using the Tensorflow/Keras machine learning pipeline, configured and trained up to 200 training iterations for a binary classification task, such as differentiating 'Normal' from 'AD' based on earlier data preprocessing steps.

Afterwards, the held-out test set was then used to assess the trained MLP model's performance. The trained model was applied to the scaled test features to generate predictions, and the probabilities results were converted into binary class labels (0 for Normal, 1 for AD) using a threshold of 0.5. Additionally, the `classification_report` and `confusion_matrix` functions in scikit-learn were used to compute the evaluation metrics, including accuracy, precision, recall, and F1-score.

1.1 Differential Methylated Regions (DMR) Analysis

In this research DMR analysis on GSE244352 was performed in order to find significant methylated regions associated with the targets APP, PSEN1, PSEN2, APOE, MAPT, TREM2. This was done by cleaning and manipulation, visualization, genomic annotation, followed by KEGG enrichment and pathway analysis of the given DMR data. This process would enable the selection of significantly methylated regions and annotating these regions with respect towards the human chromosome database, thus understanding the targeted genes affected by methylation through the specific chromosome location/regions. Lastly, these genes will be enriched and the KEGG pathway will be performed. This DMR analysis will be conducted by the use of.

Data Cleaning, Filtering, and Manipulation

From the given dataset, raw DMR data will be loaded by the use of `library(data.table)`. This allows for rapid loading of the dataset containing data on chromosome, start sequence, end sequence, p-value, q-value and methylation difference data. Thus allowing for further data manipulation. After this was done, cleaning, filtering, and transformation were carried out using `library(dplyr)`.

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This allowed for the screening of data considered significant by selecting data that met the thresholds of $q\text{-value} < 0.05$ and methylation difference > 15 . Therefore, deleting null data and enabling the collection of significantly methylated regions of the targeted genes in the chromosomes, which allow for visualization and further gene annotation.

Methylation Profiling Analysis with Manhattan Plot and Volcano Plot

After preprocessing of the data, visualization of the significant data was done by the use of library(ggplot) in order to illustrate the significance of the selected DMRs. This was done by the use of Manhattan plot to visualize the significance of each DMRs across the chromosome and Volcano plot in order to visualize the of each DMRs across the methylation differences(%). In both data, significant DMRs are annotated with the color red and non-significant DMRs are annotated with the color grey.

Genomic Annotation

Aside from visualization, annotating the DMR data is crucial in order to understand the upstream and downstream genes affected by the methylated regions. This was done by manipulating the data structure of the DMR coordinates by arranging these data as GRanges objects by the use of a library(GenomicRanges).

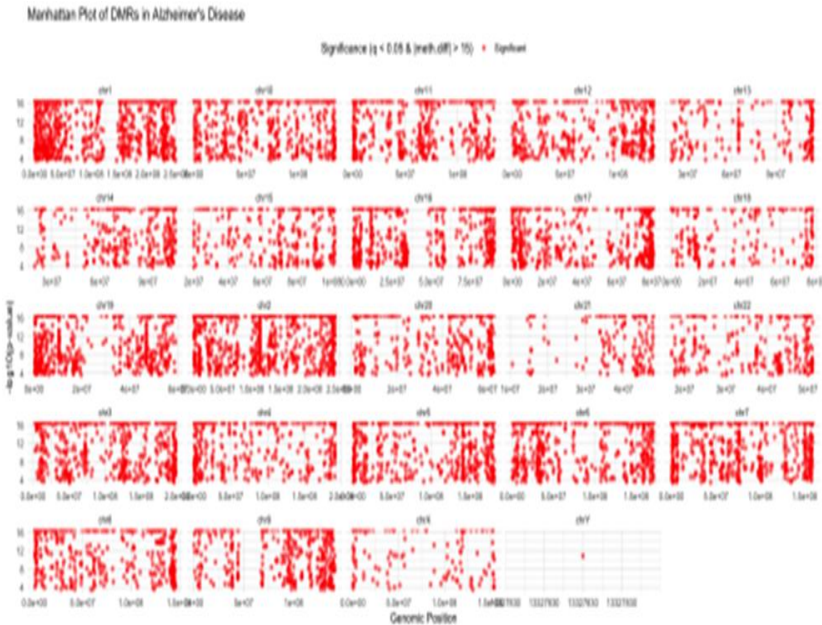
Following this, library(TxDb.Hsapiens.UCSC.hg38.knownGene) was loaded to provide a reference for human genome annotation database to map DMR based on the hg38 assembly. Furthermore, annotation is carried out by the library(ChIPseeker) as this library allows for the annotation of GRanges objects with library and library (TxDb.Hsapiens.UCSC.hg38.knownGene), thus associating the coordinates of the DMR with genomic features such as gene id, annotation (promoter, intron, exon etc), gene symbol and distance to transcription start sites (tss) which is crucial for pathway analysis. In order to allow for smooth annotation, library(rtracklayer) to ensure the compatibility of genomic ranges format during annotation.

KEGG Pathway Enrichment & Visualization

In order to conduct the KEGG pathway enrichment, the library(KEGGREST) extracted KEGG pathway data as well as genesets for the annotated gene of interest (Kanehisa et al, 2025; Kanehisa & Goto, 2000; Kanehisa, 2019). Thus, allowing for pathway visualization. After this was conducted, KEGG pathway enrichment was conducted with library(clusterProfiler), which conducted functional enrichment analysis, allowing for the identification of major pathways and subsets of related pathways affecting the pathogenesis of Alzheimer disease. With this identification, visualization of major and related pathways was conducted with a library(pathview) which visualizes the KEGG plot with reference to the methylation difference(%) on the Alzheimer disease pathway of the KEGG Pathway ID.

2. RESULTS AND DISCUSSION

Manhattan Plot of DMRs in Alzheimer Disease



Volcano Plot of DMRs in Alzheimer Disease

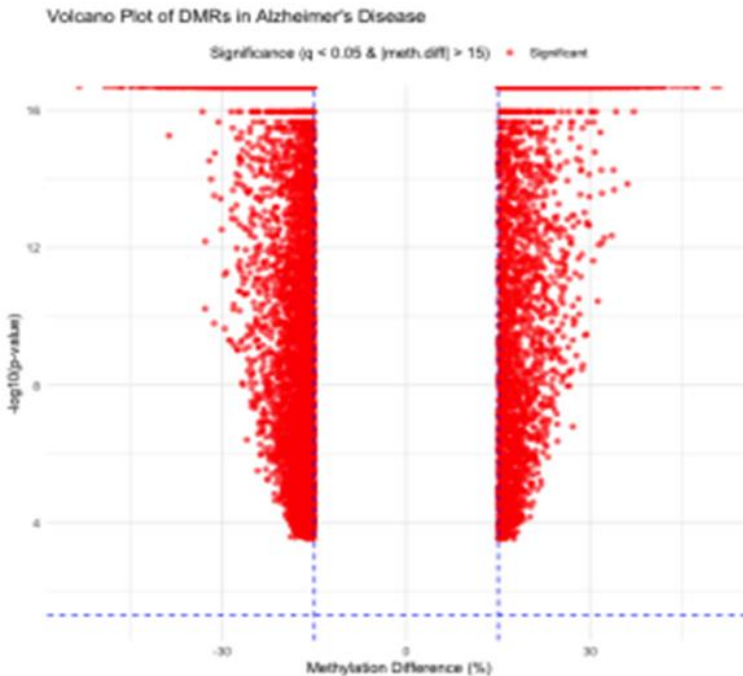


Figure 1. Combined Volcano and Manhattan Plots of differentially methylated regions (DMRs) across the genome with significance threshold of q value < 0.05 and methylation differential value > 15 .

2.1 Manhattan Plot

Figure 1 also shows a Manhattan plot that gives an example of how most of the DNA methylation changes (called DMRs) are found on chromosomes 14 and 17. The tall peaks in the plot go above the cut-off line ($q < 0.05$), showing strong signals in areas where two important Alzheimer's genes are found: PSEN1 and MAPT. Furthermore, chromosomes 13, 18, and 20 also display noticeable clusters of red dots, but lower statistical strength, indicating even though while DMRs are present, they less associated with Alzheimer pathology, one possible explanation is that these regions involves genes that have an indirect roles (e.g. immune response, synaptic function, and etc), rather than the primary amyloid/tau pathways (Feng et al., 2024).

Moreover, researchers have also studied DNA methylation across the whole genome and found that certain gene areas might be linked to Alzheimer's specifically to their pathology. For example, Monti et al. (2020) found that the PSEN1 gene had much lower methylation in Alzheimer's brains, which was linked to higher gene activity. Mori et al. (2024) also found methylation changes in the MAPT gene, suggesting this gene may not be working properly in the disease. Additionally, Gao et al. (2022) found methylation changes in both PSEN1 and MAPT genes in the front part of the brain (called the frontal cortex) in Alzheimer's patients, showing that these genes are likely involved in changes to gene regulation in the disease.

2.2 Volcano Plot

Many differentially methylated regions (DMRs) in Alzheimer's disease had high significance p-values and significant methylation differences, with several above 30% change, either rising or decreasing, as seen in the volcano plot in Figure 1. The majority of DMRs satisfy the $q < 0.05$ and methylation difference $> 15\%$ criterion, indicating a significant and pervasive epigenetic change between the control and Alzheimer's samples. To ensure that less than 5% of significant results are probable false positives, the false discovery rate is then controlled using a q-value threshold of < 0.05 (Maksimovic et al., 2016). To identify methylation variants that are biologically significant, a cutoff of greater than 15% is frequently employed; this eliminates small variations that have no effect on gene activity (Chowdhury et al., 2016). Large-scale epigenome wide association studies (EWAS), which have often found strong DNA methylation signatures linked to Alzheimer's pathology across several brain areas, are consistent with the findings. More than 334 cortical DMPs and 220 cross-cortex CpGs have been reported by meta-analyses, glial-driven epigenetic changes in neurodegeneration are demonstrated by a significant percentage of these cells originating in non-neuronal cells, including astrocytes and microglia (Shireby et al., 2022). These findings highlight the critical role of epigenetic regulation in Alzheimer's disease and suggest that specific methylation changes may serve as potential biomarkers for early diagnosis.

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Table 1. DMR identification of Alzheimer Related Genes, PSEN1 and MAPT that are significantly methylated

Chr	Start	End strand	P-value Qvalue	Meth.diff
Chr14	73113602	73113602	+0.000000e+00 0.000000e+00	-30.08647
Chr14	73198335	73198335	+0.000000e+00 0.000000e+00	19.11400
Chr17	45889839	45889839	+8.121932e-10 1.471765e-07	-24.09307

Table 2. Gene Annotations Associated with PSEN1 and MAPT

Chr	Annotation	Gene_id Gene_symbol	Distance_ to_tss
Chr14	Promoter (2-3kb)	5663 PSEN1	-2816
Chr14	Promoter (1-2kb)	5663 PSEN1	1823
Chr17	Intron (ESNT00000634876.2/100128977, intron 1 of 6)	4137 MAPT	-4688

DNA Methylation functions as an epigenetic process which controls gene expression through alterations in gene activity without modifying DNA sequences (Kiselev et al., 2021). Gene expression becomes abnormal when disruptions occur in the methylation patterns that regulate gene expression leading to neurodegenerative disease (Rasmi et al., 2022). Tables 1 and 2 show significant differences in DNA methylation with several genomic loci of PSEN1 and MAPT. Notably, the site on chromosome 14 at position 73113602 marked as hypomethylation, with a methylation difference of -30.09% (P = 0, Q = 0). On the other hand, another site on the same chromosome at position 73198335 marked as hypermethylation, showing a +19.11% increase in methylation (P = 0, Q = 0). Both regions were annotated to the promoter region of the PSEN1 gene which was located at 2.8 kb upstream and 1.8 kb downstream of the transcription start site (TSS), respectively.

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The process of methylation in promoter regions controls gene expression through silencing when hypermethylated and activation when hypomethylated (Wang et al., 2020). The opposing methylation patterns indicate complex regulatory mechanisms that could be affected by cell type and disease stage or multiple unidentified factors (Dor & Cedar, 2018).

Meanwhile, the site at 17 position 45898939 showed significant hypomethylation with -24.09% ($P = 8.12 \times 10^{-10}$, $Q = 1.47 \times 10^{-7}$) within the intronic region of the MAPT gene which is 4.7 kb upstream of the TSS. The MAPT gene produces tau protein, which has a role to stabilize microtubules in neurons. However, during Alzheimer's disease, the tau protein becomes hyperphosphorylated, and losing its ability to bind microtubules and forming the neurofibrillary tangles, this is a characteristic of Alzheimer's disease (Strang et al., 2019; Rawat et al., 2022). In addition, changes in MAPT intronic methylation patterns may impact tau transcript levels as well as tau isoform expression, both of which have been connected to tauopathy progression. The biological function of the many tau isoforms generated by MAPT alternative splicing varies because some of them are more prone to aggregation. The alterations between various tau isoforms are produced by changes in intronic methylation patterns, which could contribute to the advancement of the disease (Creekmore et al., 2024). These findings suggest that intronic methylation changes in the MAPT gene may play a direct role in modulating tau pathology in Alzheimer's disease. Targeting these epigenetic modifications could therefore offer a promising avenue for therapeutic intervention aimed at preventing or reducing tau aggregation.

Table 3. First Batch of KEGG pathway enrichment results of respective PSEN1 and MAPT with reference towards KEGG pathway ID of hsa05010 and hsa05022

ID	Description	GeneRatio	BgRatio	RichFactor	FoldEnrichment
hsa05010	Alzheimer disease	2/2	391/9396	0.005115090	24.03069
hsa05022	Alzheimer disease	2/2	483/9396	0.004140787	19.45342

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Table 4. Second Batch of KEGG pathway enrichment results of respective PSEN1 and MAPT with reference towards KEGG pathway ID of hsa05010 and hsa05022

ID	Z-Score	P-value	P.adjust	Q-value GeneID	Count
hsa05010	6.787215	0.001727434	0.01054907	0.004164108 4137/5663	2
hsa05022	6.075423	0.002637269	0.01054907	0.004164108 4137/5663	2

Using KEGG pathway enrichment analysis, Table 3 and 4 is showing the biological significance of the differentially methylated genes for PSEN1 and MAPT. The results showed both genes which correspond to KEGG IDs hsa05010 and hsa05022, were significantly linked to the Alzheimer's pathway.. The GeneRatio for both pathways is 2/2, meaning that both genes were found in each pathway. This suggests a pathway overlap with the input genes, highlighting their potential role in Alzheimer's related pathways. Furthermore, the BgRatio (Background Ratio) offers context for how frequently certain pathways occur within the gene set (Olgun et al., 2019). The BgRatio for hsa05010 is 391/9396, while for hsa05022, it is 438/9396, meaning that these pathways are connected with 391 and 483 genes, respectively, from the total of 9396 reference genes.

The rich factor values were statistically significant, with values of 0.0051 (hsa05010) and 0.0041 (hsa05022). The ratio of input genes that map a particular pathway to the total number of genes known to be in that pathway is known as the RichFactor (Hiransuchalert et al., 2024). In short, it is a measure of how dense the representation of the input genes is within a pathway. In addition, the fold enrichment values for hsa05010 and hsa05022 are 24.03 and 19.45, respectively. Fold Enrichment is calculated by comparing the GeneRatio to the BgRatio (Wu et al., 2021). In this case, both pathways are more than 19 to 24 times enriched, indicating that these genes are not just present but highly concentrated in Alzheimer's pathways. To validate the biological relevance of the observed enrichment, several statistical parameters were used: Z-score, P-value, adjusted P-value (P.adjust), and Q-value. The Z-score measures how far the observed enrichment deviates from a random distribution.

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In this analysis, the Z-scores are 6.78 (hsa05010) and 6.07 (hsa05022). These are considered statistically significant, the score above 6 represents very strong evidence, as the Z-scores above 2 are generally considered statistically significant (Andrade, 2021). The P-Values quantifies the probability of observing such enrichment under the assumption of null hypothesis . The P-values of 0.00127 and 0.00263 for the two pathways indicate that there is a 0.127% and 0.263% chance of obtaining such results by random chance, providing strong evidence of the null hypothesis (Chen et al., 2023). The adjusted P-value addresses the problem of multiple testing. Since many pathways are tested, the likelihood of false positives increases (Chicco & Agapito, 2022). Here, P.adjust values of 0.0105 for both pathways suggest that the results remain statistically significant. The Q-value on the other hand, estimates the False Discovery Rate (FDR) is 0.0041 for both pathways (Lai, 2017). This indicates that only 0.41% of the pathways found significant might actually be false positives. Q-value below 0.05 is generally accepted as high confidence (Li et al., 2021).

As seen in Figure 2. Detailed KEGG pathway illustration of hsa05010 was shown above, whereas molecular pathways and cellular effects of various mutated genes due to methylation involved in Alzheimer disease were shown in this picture. In relation to the results obtained by DMR analysis, mutation in Presenillin 1 (PSEN1) was shown to be involved in several parts of the pathways as seen in the red colored writing above. Whereas, mutations in PSEN1 was observed to affect the synthesis of γ -secretase complex, this was validated by the fact that mutations in PSEN1 functioned as a catalytic subunit in the γ -secretase which cleaved the APP (Amyloid precursor protein), thus generating amyloid beta plaques variants such as, aggregation prone (A β 42), a classic hallmark of Alzheimer's disease (Do et al., 2023). In addition, mutations in PSEN1 was also known to affect the calcium signalling pathway, promoting stress in the endoplasmic reticulum (ER), this was supported by the fact that these mutations in PSEN1 affected calcium exchange pump in the ER by binding towards the sarco/endoplasmic reticulum calcium-ATPase (SERCA) ; Furthermore, mutations in PSEN1 may also even act as a leakage channel (Deaton & Johnson, 2021).

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Lastly, these mutations in PSEN1 may impair axonal transport, this is proven as these mutations was shown to induce defects in the axonal transport of organelles and vesicles through kinesin-1 phosphorylation, which was mediated by glycogen synthase kinase 3 (GSK-3) family (Guo et al., 2020). Consequently, disrupted axonal transport can contribute to neuronal dysfunction and degeneration, exacerbating the progression of Alzheimer's disease. Understanding the molecular mechanisms behind PSEN1 mutations may provide potential targets for therapeutic strategies aimed at preserving neuronal integrity.

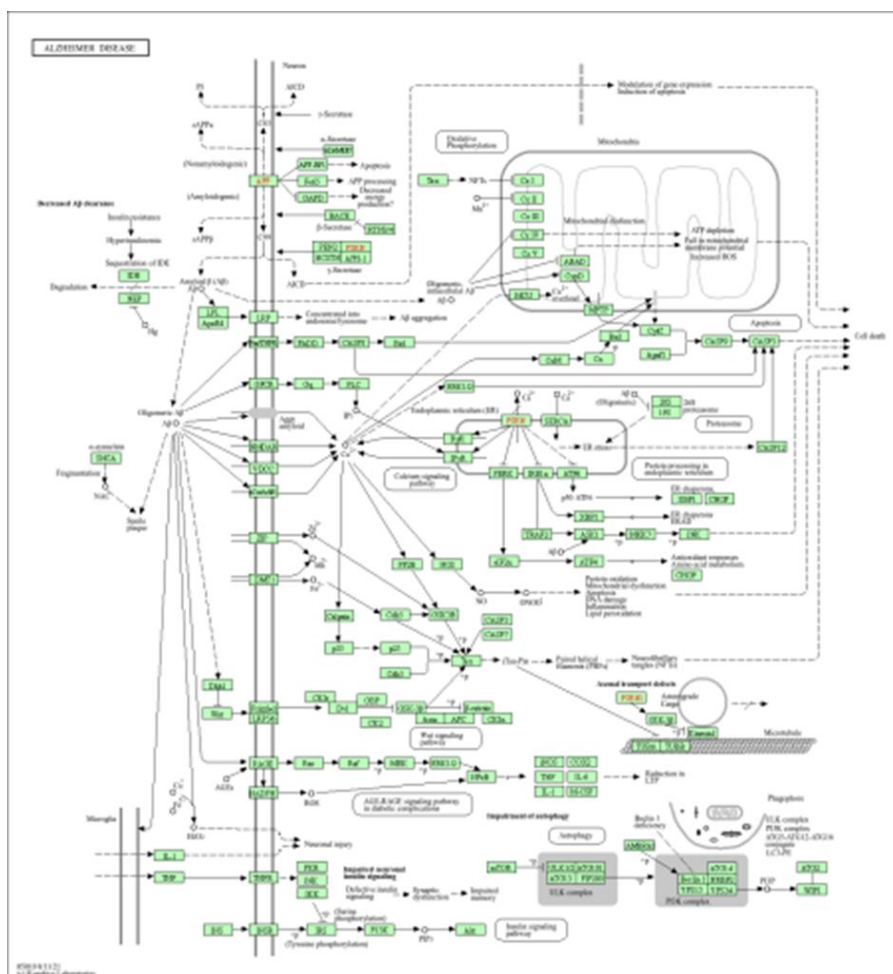


Figure 2. Alzheimer's Diseases Pathways of KEGG Pathway ID hsa05010

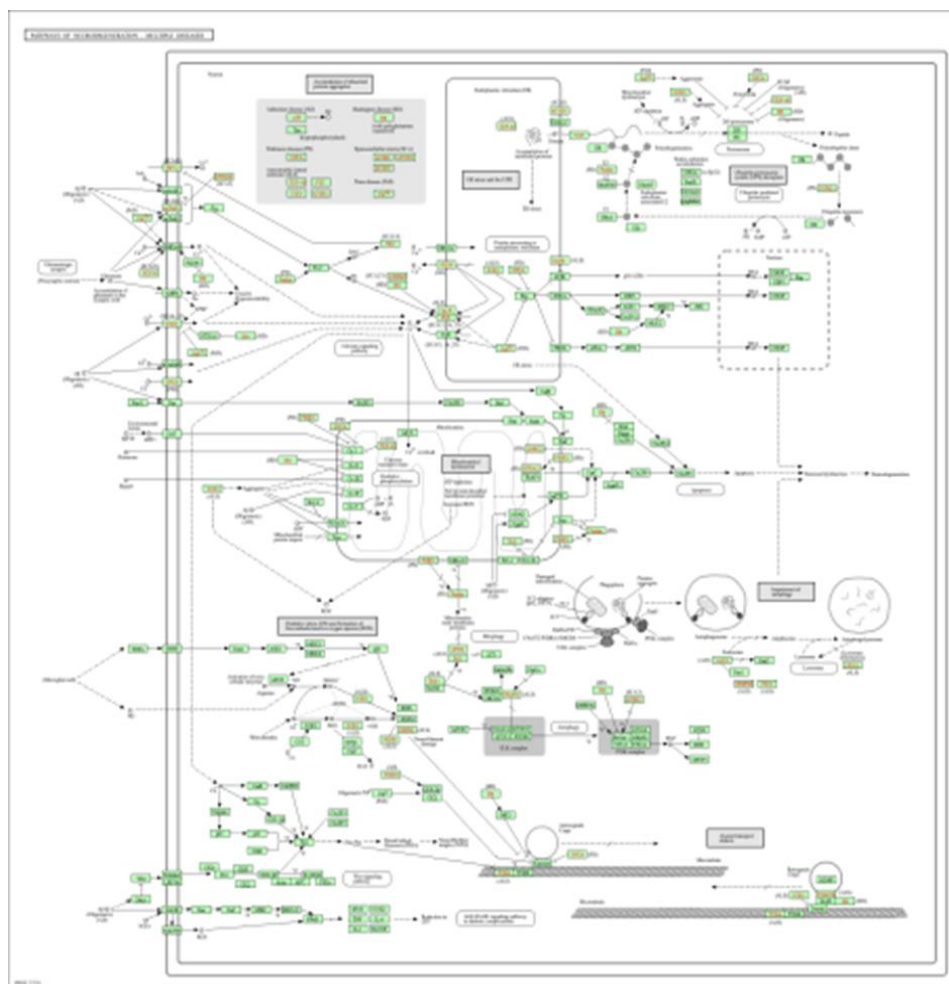


Figure 3. Alzheimer's Diseases Pathways of KEGG Pathway ID hsa05022

In Figure 3. Several critical KEGG pathways from hsa05022 influencing the pathology of several neuro-related diseases such as, Alzheimer disease, Parkinson disease, Amyotrophic lateral sclerosis, Huntington disease, Spinocerebellar ataxia (SCA) and Prion disease was shown above. Whereas, these pathways had validated the findings given in Figure 2., whereas mutations in PSEN1 was shown to affect calcium exchange in the endoplasmic reticulum through SERCA binding, disruption of axonal transport through GSK-3 β mediated kinesin phosphorylation and amyloid beta plaque formation through APP cleavage.

Further analysis of the pathway has also shown the involvement of mutations in PSEN1 in the hyperphosphorylation of tau proteins, which was a known hall mark of Alzheimer's, this was evident by the fact that mutations in PSEN1 may induce PIK3K/Akt/GSK-3 mediated signalling of tau phosphorylation, thus inducing the formation of neurofibrillary tangles (Yang et al., 2023). However in both Figure 2. And Figure 3., the significance of mutations in MAPT, which is a significant pathway observed in DMR analysis, was not shown in the KEGG pathways analysis. This was due to the fact that mutations in MAPT had only been recently proven to affect the pathology of Alzheimer's whereas both MAPT mutation variants V337M and R406W produce tau filaments with Alzheimer fold, with V337M inducing triple tau filaments in Alzheimer's disease; Thus, increasing production of neurofibrillary tangles (Qi et al., 2024). Additionally other research had noted that inconsistencies were observed in the genetic association of MAPT mutations towards late-onset Alzheimer disease and incomplete assessment of MAPT mutation variants may also affect the lack of understanding in the role of MAPT mutations in the pathogenesis of Alzheimer's disease (Juan et al., 2019). Thus, further enrichment and research must be done in order to analyze the possible critical role of MAPT mutations in Alzheimer disease.

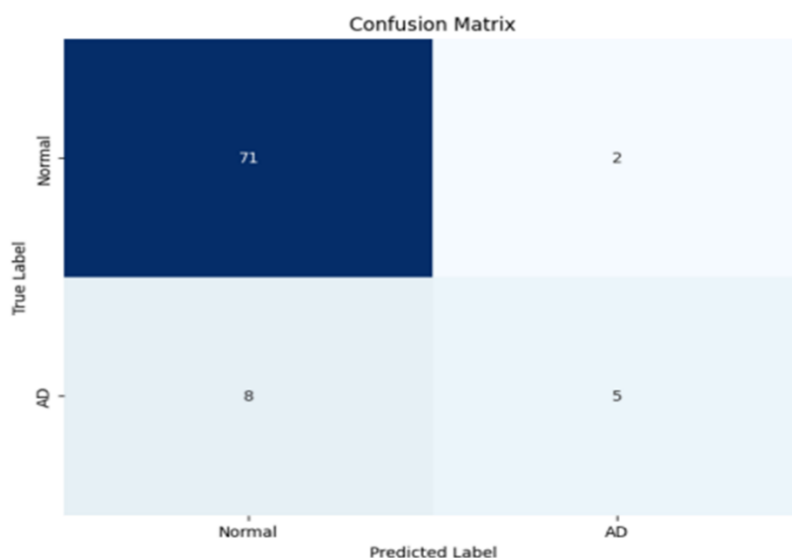


Figure 4. Machine learning Confusion Matrix

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As shown in figure 4 it shows the confusion matrix that evaluates how well the machine learning model can distinguish between people with Alzheimer's Disease (AD) and people without Alzheimer Disease, which are normal patients. It can be visualised that the matrix in the group shows the true positives and true negative values. Based on the matrix, the model did a substantial job at identifying people who are normal, as shown it correctly predicted 71 out of 73 normal cases (which are called true negatives). Only 2 normal people were mistakenly predicted as having AD (which are called false positives), showing that the model has a high specificity, and also avoiding mistakes when it comes to healthy people.

Moreover, the model was able to find some people who had Alzheimer's. Out of 13 people with AD, it correctly detected 5 of them, being described as true positives. However, it didn't detect 8 people who had Alzheimer's, and the model described it as normal (false negative). While this number is not significant, it still shows the model can recognize some signs of Alzheimer's.

One reason the model didn't perform well was because of the imbalanced information in the dataset in identifying the cases in Alzheimer's. In the data, there was an excess number of normal people when compared to Alzheimer's patients, which leads to the model learning far more about normal cases than Alzheimer's, and it can affect the decision-making and accuracy of the model itself (Albattah & Khan, 2025). Another reason that the model may predict wrong is due to the Braak stage assignment. While stages 0-2 is still too early to fulfill the clinical criteria for Alzheimer's dementia, they are still considered not normal. This factor may contribute to the positive instances mislabeled as negative, which could affect the model's performance (Rasmussen & Langerman, 2019).

CONCLUSION

Overall, this study demonstrates the complex nature of AD by combining transcriptomic analysis based on artificial intelligence with epigenetic profiling to identify important molecular signatures and predictive patterns. The study also showed how differential methylation and gene expression patterns are important in the onset and progression of AD.

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It specifically focused on genes that are strongly linked to the disease, including APP, PSEN1, APOE, MAPT, and TREM2. With the use of DMR analysis, both PSEN1 and MAPT genes have shown significant hyper and hypo-methylation patterns in Alzheimer's disease. Whereas further validation with the use of KEGG enrichment and pathway analysis have confirmed the role of these genes in molecular pathways involved in the Alzheimers such as, tau hyperphosphorylation, mitochondrial dysfunction, neuroinflammation and amyloid beta formation. There were several limitations encountered during this analysis whereas the results do not show the significance of other primary genes, thus indicating that these validated genes are less relevant. This is due to the small dataset size and lack of data variety. Thus, to solve this problem, future research needs to incorporate more datasets into the system in order to have a larger size and variance in the data. Therefore, decreasing biases towards certain genes.

In addition, the creation of a predictive model based on transcriptomic and clinical-demographic data was made possible by the complementary use of MLP. The model achieves moderate sensitivity for classifying AD and high specificity for detecting normal cases, despite certain limitations brought on by class imbalance. Although enhancements in sample balance and feature enrichment are required for clinical applicability, this implies its potential for early screening.

Finally, the integration of epigenetic knowledge with AI-driven prediction enhances the understanding of AD biology and lays the groundwork for the development of future, more complete tools. However, several limitations must be acknowledged. The class imbalance, where normal cases greatly outnumber AD cases affect the predictive model in identifying true AD cases. Additionally, although the model includes transcriptomic and clinical features, predictive accuracy could be increased by further enriching it with additional biomarkers or omics data. The findings have also not yet been verified on independent or external datasets which further limit the robustness of the findings. Moreover, the model absence of longitudinal data and real world clinical testing limits its applicability. Therefore, future studies should also concentrate on expanding the sample, verifying findings in different cohorts, incorporating longitudinal data, and examining real clinical settings.

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